Data Privacy: Homeworks #4

Due on Wednesday 11PM, June 17, 2026

Note: Submit homework via LearnUS. Topics: membership inference attacks and machine unlearning.

1. **Neyman–Pearson Lemma (Detailed Proof).** Let T be a random variable taking values in a measurable space $(\mathcal{T}, \mathcal{B})$. Consider two simple hypotheses

$$H_0 : T \sim p_0, \qquad H_1 : T \sim p_1,$$

where p_0, p_1 are densities with respect to a common dominating measure μ. A (randomized) test is a measurable function $\phi : \mathcal{T} \to [0, 1]$, where $\phi(t)$ is the probability of rejecting H_0 after observing $T = t$. Define

$$\alpha(\phi) = \mathbb{E}_{H_0}[\phi(T)] \quad \text{(size, type-I error)}, \qquad \beta(\phi) = \mathbb{E}_{H_1}[\phi(T)] \quad \text{(power)}.$$

Fix a level $\alpha_0 \in (0, 1)$. Prove the following three claims.

   (a) **Existence.** Show that there exist a threshold $\tau \geq 0$ and a randomization $\gamma \in [0, 1]$ such that the test

$$\phi^*(t) = \begin{cases} 1 & \text{if } p_1(t) > \tau\, p_0(t), \\ \gamma & \text{if } p_1(t) = \tau\, p_0(t), \\ 0 & \text{if } p_1(t) < \tau\, p_0(t), \end{cases}$$

   satisfies $\alpha(\phi^*) = \alpha_0$ exactly.
   Hint: Consider the function $G(\tau) = \Pr_{H_0}[\Lambda(T) > \tau]$ where $\Lambda(t) = p_1(t)/p_0(t)$. Show G is right-continuous and non-increasing, then pick τ so that $G(\tau) \leq \alpha_0 \leq G(\tau-)$, and choose γ to hit equality.

   (b) **Optimality.** Show that for every test ϕ with $\alpha(\phi) \leq \alpha_0$, we have $\beta(\phi) \leq \beta(\phi^*)$.
   Hint: Verify that for all t, $(\phi^*(t) - \phi(t))\,(p_1(t) - \tau\, p_0(t)) \geq 0$, then integrate.

   (c) **Uniqueness.** Suppose ϕ is any most-powerful level-α_0 test. Show that on the set $\{t : p_1(t) \neq \tau\, p_0(t)\}$ we have $\phi(t) = \phi^*(t)$ for μ-almost every t.
   Hint: The integrand in part (b) is non-negative and integrates to 0.

2. **DP $\Rightarrow$ MIA Bounds.** Let $M : \mathcal{X}^n \to \mathcal{Y}$ be (ε, δ)-differentially private, and let D_0, D_1 be neighboring datasets. Consider any (possibly randomized) binary test $\mathcal{A} : \mathcal{Y} \to \{0, 1\}$ that tries to distinguish $M(D_0)$ from $M(D_1)$. Write

$$\mathrm{FPR}(\mathcal{A}) = \Pr[\mathcal{A}(M(D_0)) = 1], \qquad \mathrm{TPR}(\mathcal{A}) = \Pr[\mathcal{A}(M(D_1)) = 1].$$

   (a) Show that

$$\mathrm{TPR}(\mathcal{A}) \leq e^{\varepsilon}\, \mathrm{FPR}(\mathcal{A}) + \delta, \qquad 1 - \mathrm{FPR}(\mathcal{A}) \leq e^{\varepsilon}\, (1 - \mathrm{TPR}(\mathcal{A})) + \delta.$$

(b) Define the MI advantage $\mathrm{Adv}(\mathcal{A}) = \mathrm{TPR}(\mathcal{A}) - \mathrm{FPR}(\mathcal{A})$. Show the loose bound

$$\mathrm{Adv}(\mathcal{A}) \le e^{\varepsilon} - 1 + \delta.$$

Then, by combining both inequalities of part (a), derive the tighter Kairouz–Oh–Viswanath bound

$$\mathrm{Adv}(\mathcal{A}) \le \frac{e^{\varepsilon} - 1 + 2\delta}{e^{\varepsilon} + 1}.$$

(c) Conclude that for $\varepsilon = 0.1$ and $\delta = 10^{-5}$, no adversary can exceed advantage ≈ 0.105, but for $\varepsilon = 8$ the bound is vacuous.

3. **Bayes-Optimal Membership Test.** Let $B \in \{0, 1\}$ be a Bernoulli(π) membership indicator and let T be an observed statistic with conditional density $p_b(t)$ given $B = b$. Define the loss of any decision rule $\hat{B} : \mathcal{T} \to \{0, 1\}$ by the 0-1 error $\Pr[\hat{B}(T) \ne B]$.

   (a) Compute the posterior $\Pr[B = 1 \mid T = t]$ in terms of p_0, p_1, π.

   (b) Show that the Bayes-optimal rule (the MAP rule) is

   $$\hat{B}_{\mathrm{MAP}}(t) = \mathbf{1}\left[\Lambda(t) \ge \frac{1-\pi}{\pi}\right], \qquad \Lambda(t) = \frac{p_1(t)}{p_0(t)},$$

   and connect this to Problem 1: the Bayes-optimal MIA is a likelihood-ratio test, only the threshold changes with the prior.

   (c) Suppose $T = \ell(f_\theta, z)$, the loss of model f_θ on the queried point z, and that $\ell \mid B = 1 \sim \mathcal{N}(\mu_{\mathrm{in}}, \sigma^2)$, $\ell \mid B = 0 \sim \mathcal{N}(\mu_{\mathrm{out}}, \sigma^2)$ with $\mu_{\mathrm{in}} < \mu_{\mathrm{out}}$. Show that the optimal rule reduces to a single-threshold loss test $\hat{B} = \mathbf{1}[\ell \le c]$ and compute c explicitly when $\pi = 1/2$.

4. **Generalization Gap as a Leakage Lower Bound (Yeom).** Let $\ell(f_\theta, z) \in [0, B]$ for all θ, z. Define

   $$R_{\mathrm{train}} = \mathbb{E}_{z \sim \mathrm{Unif}(D)}[\ell(f_\theta, z)], \qquad R_{\mathrm{pop}} = \mathbb{E}_{z \sim \mathcal{P}}[\ell(f_\theta, z)], \qquad \Delta = R_{\mathrm{pop}} - R_{\mathrm{train}} \ge 0.$$

   The MIA game: sample $b \sim \mathrm{Uniform}\{0, 1\}$. If $b = 1$, return z uniform over D; else, return $z \sim \mathcal{P}$. The adversary may use any threshold attack $\mathcal{A}_\tau(z) = \mathbf{1}[\ell(f_\theta, z) \le \tau]$. Let $F_1(\tau) = \Pr[\ell \le \tau \mid b = 1]$ and $F_0(\tau) = \Pr[\ell \le \tau \mid b = 0]$.

   (a) Show that the advantage of $\mathcal{A}_\tau$ equals

   $$\mathrm{Adv}(\mathcal{A}_\tau) = F_1(\tau) - F_0(\tau).$$

   (b) Using the layer-cake identity $\mathbb{E}[X] = \int_0^B (1 - F_X(t))\,\mathrm{d}t$ for $X \in [0, B]$, prove

   $$\int_0^B (F_1(t) - F_0(t))\,\mathrm{d}t = \Delta.$$

(c) Conclude that there exists a threshold $\tau^\star \in [0, B]$ for which the threshold attack achieves advantage

$$\mathrm{Adv}(\mathcal{A}_{\tau^\star}) \geq \frac{\Delta}{B}.$$

In other words, the generalization gap is a *lower* bound on the leakage of the best threshold attack — not an upper bound, contrary to a common informal slogan.
Hint: $\Delta = \int_0^B (F_1 - F_0)\,\mathrm{d}t \leq B \cdot \sup_t (F_1(t) - F_0(t))$.

(d) Now suppose additionally that under each hypothesis the loss is Gaussian:

$$\ell \mid b = 1 \sim \mathcal{N}(R_{\mathrm{train}}, \sigma^2), \qquad \ell \mid b = 0 \sim \mathcal{N}(R_{\mathrm{pop}}, \sigma^2).$$

Show that the maximum-advantage threshold attack has $\tau^\star = (R_{\mathrm{train}} + R_{\mathrm{pop}})/2$ and achieves

$$\mathrm{Adv}^\star = 2\Phi\left(\frac{\Delta}{2\sigma}\right) - 1.$$

For small Δ/σ, derive the linear approximation $\mathrm{Adv}^\star \approx \Delta/(\sigma\sqrt{2\pi})$ and explain why noisier loss distributions reduce leakage.

5. **LiRA Closed Form and Decision Boundary.** Assume that for a target record z, the per-shadow-model loss has Gaussian distributions under the two membership hypotheses:

$$\ell \mid (z \in D) \sim \mathcal{N}(\mu_{\mathrm{in}}, \sigma_{\mathrm{in}}^2), \qquad \ell \mid (z \notin D) \sim \mathcal{N}(\mu_{\mathrm{out}}, \sigma_{\mathrm{out}}^2).$$

(a) Derive the log-likelihood ratio

$$\log \Lambda(\ell) = \log \frac{\mathcal{N}(\ell; \mu_{\mathrm{in}}, \sigma_{\mathrm{in}}^2)}{\mathcal{N}(\ell; \mu_{\mathrm{out}}, \sigma_{\mathrm{out}}^2)} = \frac{(\ell - \mu_{\mathrm{out}})^2}{2\sigma_{\mathrm{out}}^2} - \frac{(\ell - \mu_{\mathrm{in}})^2}{2\sigma_{\mathrm{in}}^2} + \log \frac{\sigma_{\mathrm{out}}}{\sigma_{\mathrm{in}}}.$$

(b) Show that for general $\sigma_{\mathrm{in}} \neq \sigma_{\mathrm{out}}$ the optimal decision region $\{\log \Lambda > \log \tau\}$ is the complement of an interval (or a single half-line, or empty) on the loss axis. Identify when it is the complement of a bounded interval vs. a half-line.

(c) Show that in the homoscedastic case $\sigma_{\mathrm{in}} = \sigma_{\mathrm{out}} = \sigma$, the optimal MIA reduces to the single threshold $\mathcal{A}(z) = \mathbf{1}[\ell \leq c]$, and express c in terms of $\mu_{\mathrm{in}}, \mu_{\mathrm{out}}, \sigma, \tau$.

(d) In the homoscedastic case, derive the closed-form ROC: at false-positive rate α, the optimal true-positive rate is

$$\mathrm{TPR}(\alpha) = \Phi\left(\Phi^{-1}(\alpha) + \frac{\mu_{\mathrm{out}} - \mu_{\mathrm{in}}}{\sigma}\right),$$

where Φ is the standard normal CDF.

6. **Concrete MIA on a Gaussian Mean Model.** A "training" algorithm maps $D = (z_1, \dots, z_n)$ with $z_i \in \mathbb{R}$ to the parameter

$$\theta(D) = \bar{z} + W, \quad \bar{z} = \frac{1}{n}\sum_{i=1}^{n} z_i, \quad W \sim \mathcal{N}(0, \rho^2) \text{ independent.}$$

Assume the population $\mathcal{P}$ is $\mathcal{N}(0,1)$, and the adversary observes θ plus a candidate point z^*. The MIA game: $b \sim \text{Uniform}\{0,1\}$. If $b = 1$, $z^* = z_1$ (a fresh training sample, included in D); if $b = 0$, $z^* \sim \mathcal{P}$ independent of D.

   (a) Derive the conditional density of θ given z^* in each hypothesis. Show that, conditional on z^*,

$$\theta \mid (b=1) \sim \mathcal{N}\big(\tfrac{z^*}{n}, \tfrac{n-1}{n^2} + \rho^2\big), \qquad \theta \mid (b=0) \sim \mathcal{N}\big(0, \tfrac{1}{n} + \rho^2\big).$$

   (b) Show that the optimal adversary's test statistic is the (conditional) likelihood ratio $\Lambda(\theta, z^*) = p(\theta \mid b = 1, z^*)/p(\theta \mid b = 0, z^*)$, and write down $\log \Lambda$ as a quadratic in θ.

   (c) Now suppose the inputs are clipped, $|z_i| \le 1$, so the L_2-sensitivity of $\bar{z}$ under one-element neighbors is $2/n$. The Gaussian mechanism with $\rho \ge (2/n\varepsilon)\sqrt{2\log(1.25/\delta)}$ makes the release (ε, δ)-DP. Using Problem 2(b), give a numerical upper bound on the optimal MIA advantage for $\varepsilon = 0.1$, $\delta = 10^{-5}$, and comment on why $\varepsilon = 1$ already gives a vacuous bound.

7. **Influence-Function Unlearning (Closed Form).** Let $L(\theta; D) = \frac{1}{n}\sum_{i=1}^{n} \ell(\theta; z_i)$ be a twice-differentiable, λ-strongly convex loss in $\theta \in \mathbb{R}^d$ (so the Hessian $H_\theta = \nabla^2 L(\theta; D) \succeq \lambda I$). Let $\theta_o = \arg\min_\theta L(\theta; D)$ and $\theta_{-z} = \arg\min_\theta L(\theta; D \setminus \{z\})$.

   (a) Consider the parametric family $L_t(\theta) = L(\theta; D) - \frac{t}{n}\,\ell(\theta; z)$ for $t \in [0,1]$ and let $\theta(t) = \arg\min L_t(\theta)$ (so $\theta(0) = \theta_o$ and $\theta(1) = \theta_{-z}$ up to a normalization factor). Use the first-order optimality condition and the implicit function theorem to derive

$$\left.\frac{\mathrm{d}\theta(t)}{\mathrm{d}t}\right|_{t=0} = \frac{1}{n}\, H_{\theta_o}^{-1}\, \nabla_\theta \ell(\theta_o; z).$$

   (b) Conclude the first-order Newton-step approximation

$$\theta_{-z} - \theta_o \approx \frac{1}{n}\, H_{\theta_o}^{-1}\, \nabla_\theta \ell(\theta_o; z).$$

   Hence the "unlearned" parameter is $\theta_u = \theta_o + \frac{1}{n} H_{\theta_o}^{-1} \nabla_\theta \ell(\theta_o; z)$.

   (c) Assume in addition that ℓ is G-Lipschitz in θ (so $\|\nabla_\theta \ell\| \le G$). Show that the L_2-sensitivity of the unlearned parameter to the choice of deleted point z versus its neighbor z' is bounded:

$$\sup_{z,z'} \|\theta_u(D, z) - \theta_u(D, z')\| \le \frac{2G}{\lambda n}.$$

   (Treat $H_{\theta_o}^{-1}$ as fixed with operator norm $\le 1/\lambda$; this is the first-order sensitivity used in certified unlearning.)

8. **Certified (ε, δ)-Unlearning by Gaussian Noise.** Building on Problem 7, let $A(D) = \arg\min_\theta L(\theta; D)$ be the (deterministic) retrain map, and let

$$\mathcal{U}(D, z) = \theta_o(D) + \frac{1}{n} H^{-1}_{\theta_o(D)} \nabla_\theta \ell(\theta_o(D); z) + N, \qquad N \sim \mathcal{N}(0, \sigma^2 I_d).$$

Assume ℓ is λ-strongly convex and G-Lipschitz, and that the first-order approximation in Problem 7(b) has L_2-error bounded by $\eta := \|\theta_u^{\text{1st-order}} - A(D \setminus \{z\})\| \le C_2 G^2/(\lambda^2 n^2)$ for some universal constant C_2 (a known second-order bound; you may use this without proof).

(a) Show that the total L_2-sensitivity (relative to retrain) is bounded:

$$\Big\| \mathcal{U}(D, z) - N - A(D \setminus \{z\}) \Big\| \le \eta \le \frac{C_2\, G^2}{\lambda^2 n^2}.$$

(b) Apply the Gaussian mechanism privacy bound: if $\sigma \ge \eta \sqrt{2 \log(1.25/\delta)}/\varepsilon$, then $\mathcal{U}(D, z)$ is (ε, δ)-indistinguishable from $A(D \setminus \{z\}) + N$. Conclude

$$\sigma \ge \frac{C_2\, G^2}{\lambda^2 n^2 \varepsilon} \sqrt{2 \log(1.25/\delta)}$$

gives an (ε, δ)-certified unlearning algorithm.

(c) How does the required noise scale with n? Compare to the Gaussian-mechanism noise needed for (ε, δ)-DP of the trained model itself ($\sigma \propto 1/(n\varepsilon)$). Explain in one or two sentences why the unlearning noise is smaller for large n.

9. **NPO Has a Bounded Gradient; GA Does Not.** For LLM unlearning, let $\pi_\theta(y \mid x)$ denote the model's likelihood of completion y given prompt x, and let π_{ref} be a fixed reference model. The forget set is $\mathcal{D}_F$. Two unlearning losses:

$$\mathcal{L}_{\text{GA}}(\theta) = -\,\mathbb{E}_{(x,y)\sim\mathcal{D}_F}[-\log \pi_\theta(y \mid x)] = \mathbb{E}[\log \pi_\theta(y \mid x)]\,,$$
$$\mathcal{L}_{\text{NPO}}(\theta) = \tfrac{2}{\beta}\, \mathbb{E}\Big[\log\Big(1 + \big(\tfrac{\pi_\theta(y|x)}{\pi_{\text{ref}}(y|x)}\big)^\beta\Big)\Big] \quad (\beta > 0).$$

(a) Compute $\nabla_\theta \mathcal{L}_{\text{GA}}$. Using the identity $\nabla_\theta \log \pi_\theta = \nabla_\theta \pi_\theta / \pi_\theta$, show that for any sample with $\pi_\theta(y \mid x) \to 0$ (and $\|\nabla_\theta \pi_\theta\|$ bounded away from zero in some direction), $\|\nabla_\theta \mathcal{L}_{\text{GA}}\| \to \infty$. Explain in one sentence why this drives the GA optimization to escape the data distribution ("GA collapse").

(b) Let $u(\theta) = \beta \log(\pi_\theta(y \mid x)/\pi_{\text{ref}}(y \mid x))$. Show that

$$\mathcal{L}_{\text{NPO}}(\theta) = \frac{2}{\beta}\, \mathbb{E}[\log(1 + e^{u(\theta)})], \qquad \frac{\partial \mathcal{L}_{\text{NPO}}}{\partial u} = \frac{2}{\beta}\, \sigma(u) \in \Big(0, \tfrac{2}{\beta}\Big)\,,$$

where σ is the logistic sigmoid. Conclude that

$$\big\|\nabla_\theta \mathcal{L}_{\text{NPO}}\big\| \le 2\, \mathbb{E}\big[\|\nabla_\theta \log \pi_\theta(y \mid x)\|\big].$$

(c) Show that this gradient bound is uniform in θ (it does not blow up as $\pi_\theta \to 0$), in sharp contrast to part (a). State briefly why this bounded-per-step divergence is what makes NPO "safe" for unlearning.

10. **Black-Box Unlearning Audits Require Many Queries.** Let θ_o and θ_u be two LLMs with output distributions P_0 and P_1 on prompts $x \in \mathcal{X}$. Let $\mathrm{TV}(P_0, P_1) = \frac{1}{2}\int |p_0(y \mid x) - p_1(y \mid x)| \, \mathrm{d}\mu(y, x) \le \eta$ (averaged over a fixed query distribution). An auditor draws q i.i.d. prompts $x_1, \ldots, x_q$, observes one sample $y_i \sim P_b(\cdot \mid x_i)$ per prompt, and outputs $\hat{b} \in \{0, 1\}$.

   (a) **(Le Cam two-point lower bound.)** Show that for any test based on q samples,

   $$\Pr_{b=0}[\hat{b} \neq 0] + \Pr_{b=1}[\hat{b} \neq 1] \ge 1 - \mathrm{TV}(P_0^{\otimes q}, P_1^{\otimes q}).$$

   Hint: For any test ψ and any pair of distributions P, Q, $\mathbb{E}_P[\psi] - \mathbb{E}_Q[\psi] \le \mathrm{TV}(P, Q)$.

   (b) Show that $\mathrm{TV}(P_0^{\otimes q}, P_1^{\otimes q}) \le q \cdot \mathrm{TV}(P_0, P_1) \le q\eta$ by a hybrid (telescoping) argument over the q coordinates.

   (c) Conclude that to guarantee total error $\le \alpha$ (i.e., $\Pr_0[\hat{b} \neq 0] + \Pr_1[\hat{b} \neq 1] \le 2\alpha$), the auditor needs

   $$q \ge \frac{1 - 2\alpha}{\eta}.$$

   Tighter χ^2-style bounds give $q = \Omega(1/\eta^2)$. Why does this make single-prompt audits of LLM unlearning fundamentally insufficient when θ_u, θ_o differ only by a small distributional perturbation?